Abhinav Avasarala

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Education

North Carolina State University (NCSU) **May 2027 (Expected)**
B.S. Computer Engineering *Raleigh, NC*

- **GPA:** 4.00/4.00
- **Courses**: Architecture of Parallel Computers, Operating Systems, Data Structures & Algorithms, Machine Learning, Digital Design, C & Software Tools, HPC & Apache Spark

Technical Skills

Languages: Python, C++, C, Java, SQL, Verilog, x86 Assembly, Bash/Shell
GPU & HPC: CUDA, cuBLAS, Triton, cuDNN, PyTorch, Nsight Systems, Nsight Compute, MPI, OpenMP
Schedulers & Orchestration: Slurm, LSF, Kubernetes, Docker, Apptainer, GKE
ML & Systems: PyTorch, HuggingFace, low-precision inference (INT8/FP8), kernel profiling, RAG
Tools: Git, Linux/Unix, Prometheus, FastAPI, PostgreSQL, NumPy, Pandas

Experience

NCSU Office of Information Technology **May 2026 – Present**
HPC Systems Intern | Slurm, LSF, Linux, Python, MPI *Raleigh, NC*

- Supporting enterprise-scale migration of workload scheduling from IBM LSF to Slurm across the **Hazel HPC cluster (1,000+ nodes)**, analyzing job submission patterns and translating batch scripts for production rollout
- Built **Apptainer** container images for GPU-accelerated PyTorch workloads, packaging CUDA/cuDNN dependencies for reproducible, rootless execution across heterogeneous compute nodes
- Characterized multi-GPU interconnect topology on GPU nodes, identifying cross-NUMA GPU placement and hybrid **InfiniBand/RoCE** network paths, and analyzed the resulting bandwidth implications for **NCCL** collective operations
- Automated strong-scaling benchmarks and MPI rank-distribution studies with Python and Slurm job arrays, producing throughput reports that validate cluster performance across node configurations

NCSU Spiking Neural Network Research **Jan 2026 – May 2026**
GPU Research Intern, Advisor: Dr. Huiyang Zhou | CUDA, Triton, PyTorch, Nsight Compute *Raleigh, NC*

- Profiled and optimized a Triton-based LIF spiking-neuron kernel (SpikingJelly) on NVIDIA **H100/H200** GPUs, achieving a ∼**4.45x throughput improvement** over the PyTorch baseline
- Analyzed kernel efficiency, memory bandwidth, and occupancy with **Nsight Compute**; iterated on launch configurations and resolved on-cluster profiling permissions to raise GPU utilization
- Benchmarked spiking-network inference efficiency across GPU hardware, contributing results toward a Triton kernel publication

iQuadra Information Services **May 2025 – August 2025**
Software Engineer Intern | ReactJS, Node.js, PostgreSQL, REST APIs *Remote*

- Built a full-stack bug-tracking system (ReactJS, Node/Express, PostgreSQL) with **12+ RESTful endpoints** for end-to-end issue lifecycle management, improving bug-resolution time ∼**20%** via priority-tagging workflows
- Led a team of 3 front-end developers building responsive UIs from Figma mockups; coordinated agile sprints and code reviews

Projects

KubeMind: Natural-Language Kubernetes Agent | *Python, LangGraph, Kubernetes, GKE, Prometheus*

- Built a natural-language Kubernetes CLI agent that automates node-pool provisioning, deployment scaling, and cluster diagnostics via single English commands across GKE infrastructure
- Integrated **Prometheus** for natural-language health queries, surfacing crash-loops, CPU spikes, and anomalies on pass/warn/critical dashboards; built deterministic undo across 7 op types using state snapshots
- Implemented a 3-layer safety system (NeMo Guardrails + deterministic risk classifier) intercepting dangerous ops; benchmarked model planning across 19 labeled test cases with precision/recall/F1 (**89% pass rate**)

Stock Sentiment Analyzer: Financial NLP Pipeline | *Python, FastAPI, FinBERT, HuggingFace*

- Built an end-to-end ML pipeline where users input portfolios and receive real-time news sentiment on holdings, aggregating sources from Reddit, Yahoo Finance, and NewsAPI
- Compared FinBERT and VADER and selected FinBERT for **92% test accuracy** on financial-text sentiment classification

CUDA CNN Ops From Scratch: GPU Kernel Optimization | *CUDA, C++, cuDNN, Nsight Compute*

- Implemented core CNN operators (2D convolution, ReLU, max-pool) in CUDA and validated correctness against PyTorch across **100+ randomized test cases**
- Optimized the baseline convolution kernel with shared-memory tiling and improved memory coalescing for a **2x speedup** (Nsight Compute); benchmarked against cuDNN to identify key performance drivers